US012402618B2

(12) **United States Patent**
Rubel et al.

(10) **Patent No.: US 12,402,618 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **INSECT ELIMINATING DEVICE WITH SIMULATED FLAME AND VACUUM SUCTION ELEMENT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/679,913**

(22) Filed: **May 31, 2024**

(65) **Prior Publication Data**

US 2025/0057142 A1 Feb. 20, 2025

Related U.S. Application Data

(60) Provisional application No. 63/533,243, filed on Aug. 17, 2023.

(51) **Int. Cl.**
A01M 1/08 (2006.01)

(52) **U.S. Cl.**
CPC ***A01M 1/08*** (2013.01)

(58) **Field of Classification Search**
CPC A01M 1/08
See application file for complete search history.

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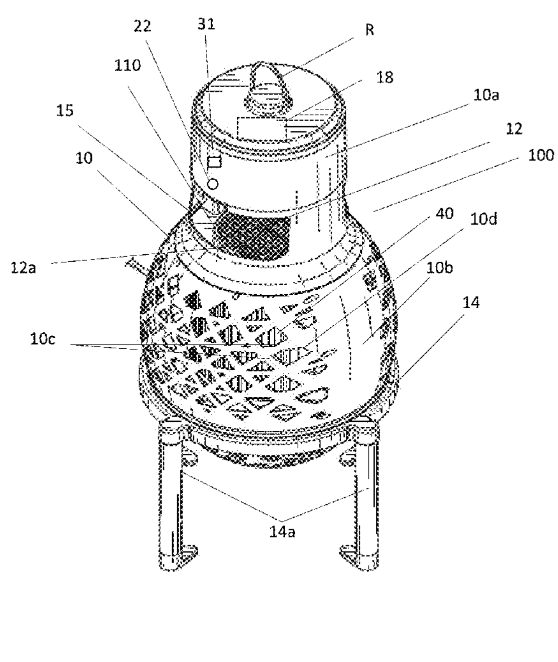
(57) **ABSTRACT**

An insect eliminating device includes a body including a top portion and a bottom portion provided below the top portion; a light portion mounted in the body and including: a UV light element mounted in an open space provided in the top portion; a flickering light portion mounted in the bottom portion; a central conduit providing fluid communication between the open space in the top portion and the bottom portion of the body; a fan, mounted in the bottom portion to direct air from the open space in the top portion, through the central conduit and into the lower portion; and a grid mounted at a bottom end of the central conduit including a plurality of openings sized to allow air and insects to pass downward into the bottom portion of the base and prevent insects from passing upwards into the central conduit.

20 Claims, 4 Drawing Sheets



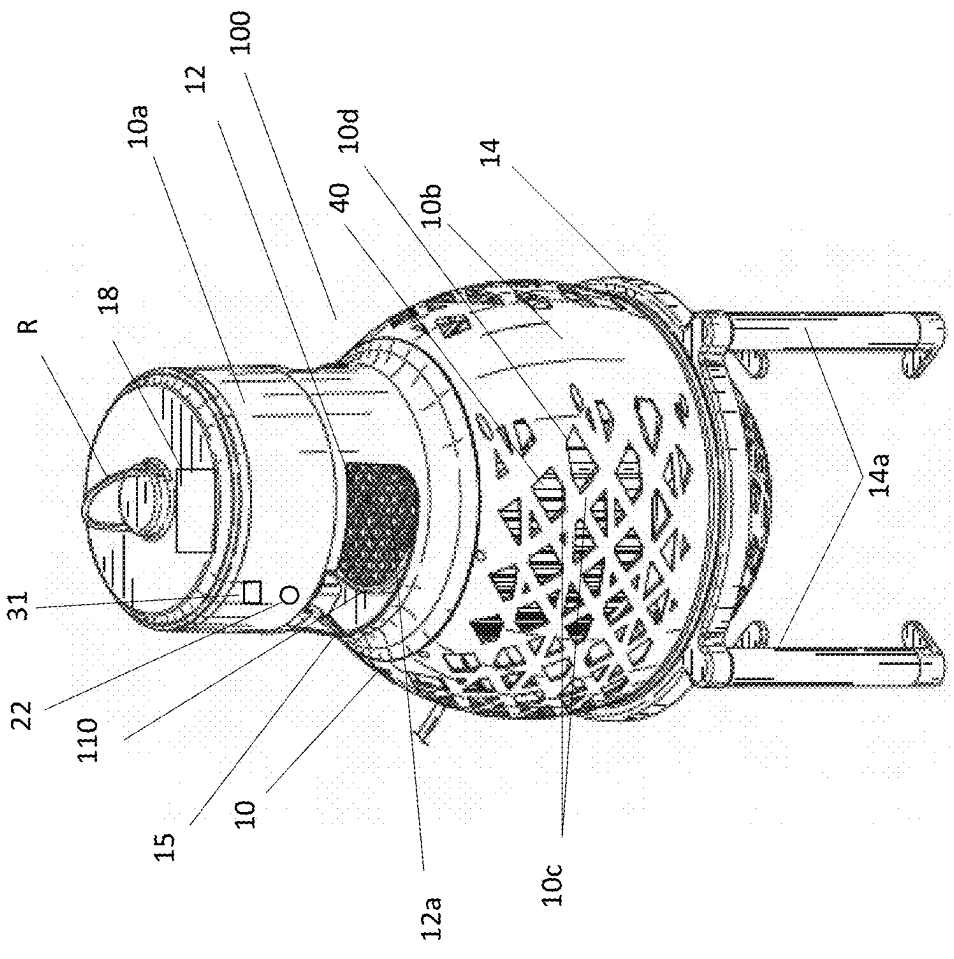


FIG. 1

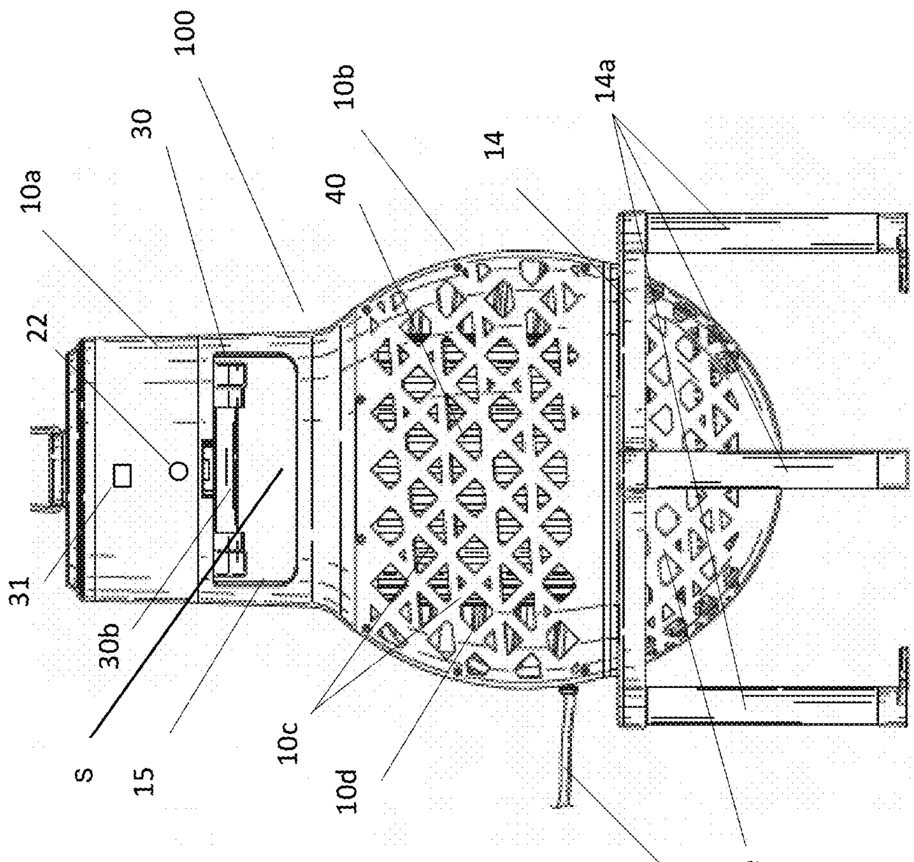


FIG. 2

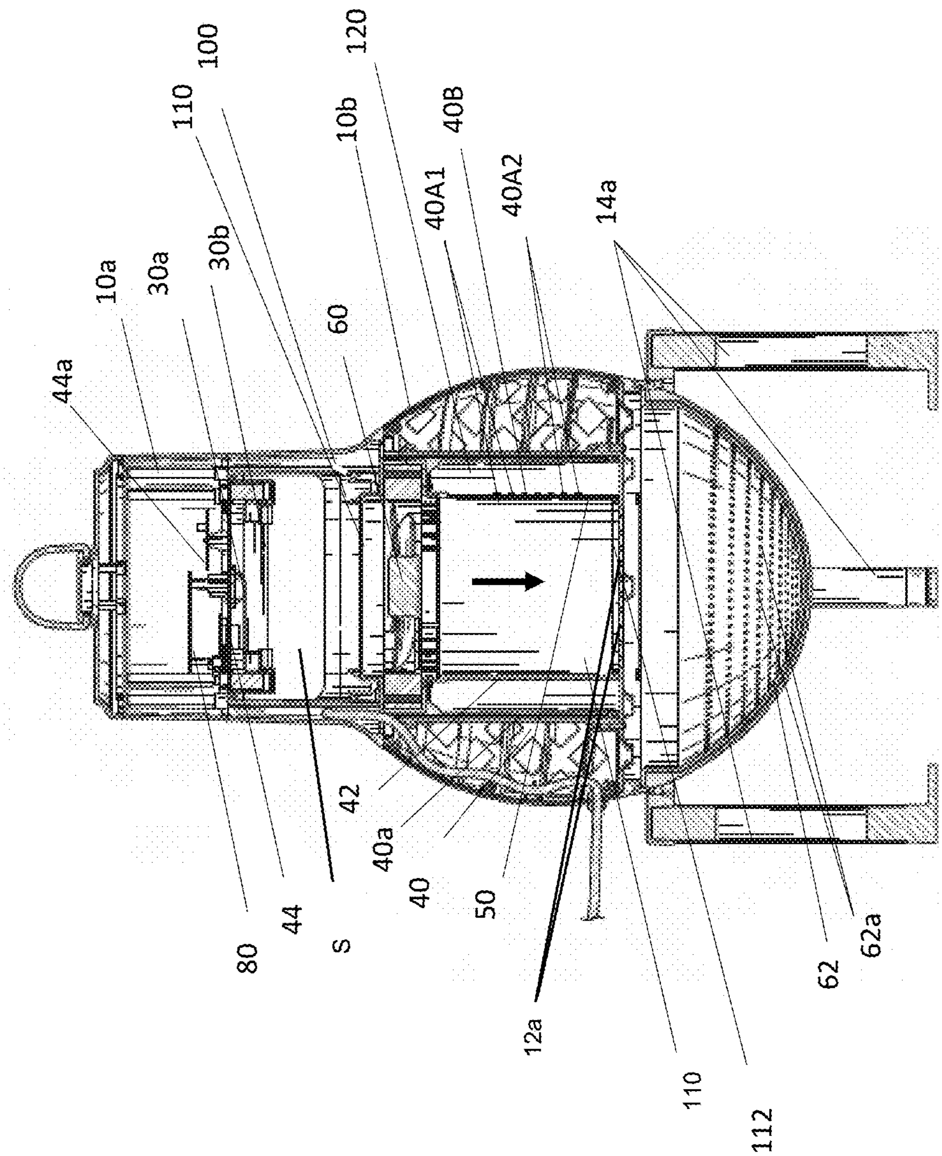

44a
10a
30a
30b
110
100
60
10b
120
40A1
40B
40A2
14a
80
44
S
42
40a
40
50
12a
110
112
62
62a

FIG. 3

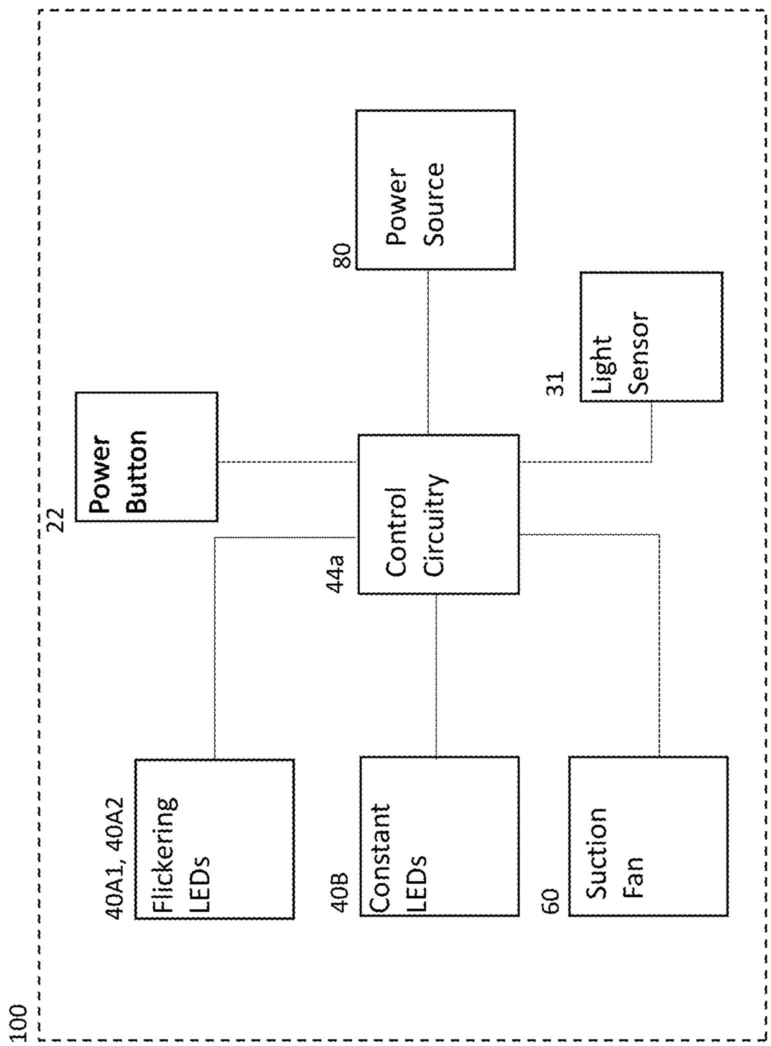

100
22
Power Button
80
Power Source
31
Light Sensor
44a
Control Circuitry
40A1, 40A2
Flickering LEDs
40B
Constant LEDs
60
Suction Fan

FIG. 4

INSECT ELIMINATING DEVICE WITH SIMULATED FLAME AND VACUUM SUCTION ELEMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims benefit of and priority to U.S. provisional patent application Ser. No. 63/533,243 filed Aug. 17, 2023 entitled INSECT ELIMINATING DEVICE WITH SIMULATED FLAME AND VACUUM SUCTION ELEMENT, the entire content of which is hereby incorporate by reference herein.

BACKGROUND

Field of the Disclosure

The present invention relates to an insect eliminating device with a lighting element and a suction element to draw insects into the device.

Related Art

Conventional electric insect eliminating devices are typically focused on pest control and include features that optimize pest removal, however, ignore aesthetics and other useful functionality. While conventional electric insect eliminating devices often emit UV light to attract insects, they do not provide sufficient light to aid those around them to see. Further, their design is typically not aesthetically pleasing.

Further, conventional light based insect eliminating devices typically use an electrified grid to eliminate insects. While such devices may be effective, they rely on the unpredictable activity of the insect to contact the electrified grid in order to eliminate the insect.

Accordingly, it would be beneficial to provide an insect eliminator that avoids these and other problems.

SUMMARY

It is an object of the present disclosure to provide an insect eliminating device including a light portion that provides a simulated flame as well as a suction element to draw insects into the device.

An electric insect eliminating device in accordance with an embodiment of the present disclosure includes: a body including a top portion and a bottom portion provided below the top portion; a light portion mounted in the body and including: a UV light element mounted in an open space provided in the top portion; a flickering light portion mounted in the bottom portion; a central conduit providing fluid communication between the open space in the top portion and the bottom portion of the body; a fan, mounted in the bottom portion to direct air from the open space in the top portion, through the central conduit and into the lower portion; and a grid mounted at a bottom end of the central conduit including a plurality of openings sized to allow air and insects to pass downward into the bottom portion of the base and prevent insects from passing upwards into the central conduit.

In embodiments, the UV light element is a UV light emitting diode.

In embodiments, the flickering light portion includes a first group of light emitting elements configured to emit light in a first pattern and a second group of light emitting elements configured to emit light in a second pattern.

In embodiments, the insect eliminating device includes a support provided in the bottom portion of the body, wherein the first group of light emitting elements and the second group of light emitting elements are mounted on the support.

In embodiments, the insect eliminating device includes a shade mounted in the bottom portions of the body, wherein the shade surrounds the flickering light portion.

In embodiments, the light portion includes at least one constant light element.

In embodiments, the insect eliminating device includes a cage element provided around the bottom portion of the body configured to prevent insects from exiting the bottom portion of the body.

In embodiments, the insect eliminating device includes a control circuit operably connected to the light portion and the fan to selectively activate the UV light portion, the flickering light portion and the fan to draw insects from the open space in the top portion into the central conduit and to the bottom portion of the body.

In embodiments, the insect eliminating device includes a light sensor operably connected to the control circuit, wherein the control circuit selectively activates the light portion and the fan based on an ambient light level detected by the light sensor.

In embodiments, the insect eliminating device includes an input device operably connected to the control circuit, wherein the control circuit selectively activates the light portion and the fan based on input provided by the input device.

In embodiments, the insect eliminating device includes a power source mounted in the body and configured to provide power to the control circuit, light portion and the fan.

In embodiments, the power source is a battery.

In embodiments, the power source is a rechargeable battery.

In embodiments, the insect eliminating device includes a solar panel mounted on a top of the body and operably connected to the power source to recharge the power source.

In embodiments, the control circuit includes a charging circuit configured to control charging of the power source.

In embodiments, the insect eliminating device includes a wire connectable to an exterior power source, wherein the wire is electrically connected to the power source and operable to recharge the power source.

In embodiments, the insect eliminating device includes an input device operably connected to the control circuit, wherein the control circuit selectively operates the light portion and the fan based on input provided by the input device.

In embodiments, the insect eliminating device includes a stand configured to receive the body.

In embodiments, the light portion includes a constant light source.

In embodiments, the constant light source includes a plurality of constant light elements.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and related objects, features and advantages of the present disclosure will be more fully understood by reference to the following, detailed description of the preferred, albeit illustrative, embodiments of the present invention when taken in conjunction with the accompanying figures, wherein:

FIG. **1** illustrates a front perspective view of an insect eliminating device in accordance with an embodiment of the present disclosure;

FIG. **2** illustrates a front view of the insect eliminating device of FIG. **1** in accordance with an embodiment of the present disclosure;

FIG. **3** illustrates a cross-sectional view of the insect eliminating device of FIGS. **1-2** in accordance with an embodiment of the present disclosure;

FIG. **4** illustrates an exemplary block diagram of the insect eliminating device of FIGS. **1-4**.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

An insect eliminating device **100** in accordance with an embodiment of the present disclosure is illustrated in FIG. **1**. In embodiments, the insect eliminating device **100** may include a body **10** including a top portion **10***a* and a lower portion **10***b*. In embodiments, the body **10** may be mounted in a stand **14** including a plurality of legs **14***a*. In embodiments, the stand **14** may not be used. In embodiments, the lower portion **10***b* of the body **10** may include a plurality of ribs **10***c* which may be arranged in a decorative pattern such that openings **10***d* are provided between them. In embodiments, a ring R or other mounting element may be provided on a top surface of the top portion **10***a* suitable for hanging or otherwise supporting the device **100**.

In embodiments, a light portion **40** may be mounted in the interior of the lower portion **10***b* of the body **10**. In embodiments, the light portion **40** may be visible through the openings **10***d* between the ribs **10***c* in the lower portion **10***b* of the body **10**. In embodiments, the light portion **40** may include a shade **40***a* surrounding a plurality of light sources **40A1**, **40A2** (see FIG. **3**, for example). In embodiments, the shade **40***a* may be colored or may include multiple colors. In embodiments, the shade **40***a* may be substantially translucent. In embodiments, the light sources **40A1**, **40A2** may be mounted on a flexible printed circuit board **42**. In embodiments, the printed circuit board **42** may be wrapped around a support element **50**. In embodiments, the support element **50** may be cylindrical in shape and the printed circuit board **42** may be wrapped around the support. In embodiments, the support element may not be included. In embodiments, the light elements **40A1**, **40A2** may be mounted directly on the support element **50**. In embodiments, the light sources **40A1**, **40A2** are light emitting diodes (LEDs). In embodiments, the light sources **40A1**, **40A2** may be white light LEDs that emit light through the shade **40***a* which may be tinted or colored to provide the impression of a flame. In embodiments, one or more of the LEDs **40A1**, **40A2** may be a different color in order to provide the impression of a flame. In embodiments, the LEDs **40A1** belong to a first group and the LEDs **40A2** belong to a second group. In embodiments, the first group of the LEDs **40A1** may be driven to blink on and off together. In embodiments, the second group of LEDs **40A2** may be driven to brighten and dim in intensity together. In embodiments, the second group of LED's **40A2** may be driven to blink on and off together and the first group of LED's **40A1** may be driven to brighten and dim in intensity together. In embodiments, the second group of LEDs **40A2** may be positioned below the first group of LEDs **40A1** or vice versa. In embodiments the first group of LED's **40A1** may be driven in a first pattern and the second group of LED's **40A2** may be driven in a second pattern. In embodiments, one or more of the light emitting diodes in the first group of LEDs **40A1** or the second group of **40A2** may be of different colors. In embodiments, the light elements **40A1**, **40A2** may be any suitable light source. The combination of the two groups of LEDs, when viewed through the shade **40***a*, which may be tinted or colored as noted above, if desired, provides the impression of a flickering flame. In embodiments, the shade **40***a* may be of any desired shape provided that it fits within the body **10***b*.

In embodiments, an opening **15** may be provided in a top portion **10***a* of the body **10**. In embodiments, the opening **15** passes from one side of the top portion **10***a* to the other as is generally indicated in FIG. **2**, for example. In embodiments, the opening **15** provides access to an interior of the top portion **10***a* of the body **10** from opposed sides thereof. In embodiments, a central conduit **110** provides fluid communication between a central portion **120** of the lower portion **10***b* of the body **10** and the open area of the top portion **10***a* of the body **12**. In embodiments, a grate **12** including openings **12***a* that are sized to allow insects to pass therethrough is provided at a bottom of the central conduit **110**.

In embodiments, a UV light source **30** may be mounted in the top portion **10***a* of the body **10**. In embodiments, the UV light source **30** is visible to insects outside of the body **10** and attracts them through the opening **15** and into the top portion **10***a* of the body **10**. In embodiments, the UV light source **30** may include a light emitting element **30***a* as well as a cover or lens **30***b* (see FIG. **3**). In embodiments, the light emitting element **30***a* may be a light emitting diode (LED) or any other suitable light source. In embodiments, the lens **30***b* may be a UV lens that may be used to emit UV light to attract the insects. In embodiments, the light emitting element **30***a* may emit UV light without the need for the lens **30***b*.

In embodiments, the lower portion **10***b* of the body **10** may include a suction fan **60** mounted in fluid communication with the central conduit **110**. In embodiments, the suction fan **60** may be selectively operable to draw air into the lower portion **10***b*, including any insects that have passed through the opening **15** into the interior open space S of the top portion **10***a*. In embodiments, the fan **60** may draw air and insects through a top grate **12** positioned at a top end of a central conduit **110**. In embodiments, an interior of the support **50** may form the central conduit **110** through which the air and insects may be drawn. In embodiments, the central conduit **110** provides for a seal between the exterior of the body **12** and the interior such that the operation of the fan **60** draws insects into the conduit and down into the base **10***e*. In embodiments, where the support **50** is not used, the flexible PCB **42** may form the central conduit **110**. In embodiments, the base **10***e* of the lower portion **10***b* of the body **10** may include a net or mesh covering **62** including openings **62***a* to allow air to pass but prevent insects from escaping. In embodiments, the net or mesh **62** may be provided only in the base **10***e* of the lower portion **10***b* of the body **10** and may be separated from the top portion of the lower portion **10***b* via grate **112** positioned at a lower end of the conduit **110** that allows insects to pass through but prevents them from moving back up. In embodiments, the net or mesh **62** is only provided in the base **10***c* such that it does not interfere with the aesthetic appearance of the ribs **10***c* or the view of the lighting element **40** from the exterior of the body **12**. In embodiments, the openings **12***a* in the grate **112** are large enough to allow the insects to pass through but sufficiently small to prevent them from flying out.

In embodiments, the PCB **42** may include additional constant light sources **40**B that may be used to provide a constant light rather than the flickering effect discussed above. In embodiments, the constant light sources **40**B may be of the same color and evenly spaced around the substrate **42**. In embodiments, the light sources **40**B may be activated together and remain on at the same time to provide for a constant light that does not flicker. In embodiments, the light sources **40**B may be white light LEDs. In embodiments, when the light sources **40**B are activated, the light sources **40**A**1**, **40**A**2** are not activated such that the light portion **10** may operate in a constant light mode or a flickering mode. In embodiments, the LEDs **40**A**1**, **40**A**2** may be used to provide constant light by simply being operated in an on state without blinking or brightening and dimming. In embodiments, where the LEDs **40**A**1**, **40**A**2** are used to provide constant light, it may not be necessary to provide the light sources **40**B.

In embodiments, a circuit board **44** (see FIG. **3**) may be provided to hold or provide control circuitry **44***a* for the lighting element **40**, the UV light **30** as well as the fan **60**. In embodiments, a power source **80** may be operatively connected to the PCB **44**. In embodiments, the power source **80** may be provided on the circuit board **44**. In embodiments, the control circuitry **44***a* may be provided elsewhere and need not be mounted on the circuit board **44**. In embodiments, the power source **80** may be mounted elsewhere. In embodiments, the power source **80** may be a transformer or a capacitor and may be connected to an external voltage source via the wire W. In embodiments, the power source **80** may be a rechargeable battery. In embodiments, the rechargeable battery may be recharged via a line voltage via wire W or any other suitable connection, for example, via a USB, wireless or other connection. In embodiments, the power source **80** may be recharged via any other suitable power source. Any suitable power source **80** may be used. In embodiments, a solar panel may be provided and may be used to provide power directly or to recharge the rechargeable battery **80**.

In embodiments, the control circuitry **44***a* may include an LED boost circuit (or control circuit) used to drive the UV LED **30***a* and/or the LEDs **40**A**1**, **40**A**2** and/or LEDs **40**B as well. In embodiments, other driving circuitry may be provided to drive the UV LED **30***a* and/or the LEDs **40**A**1**, **40**A**2** and **40**B. As noted above, the two groups of LEDs **40**A**1**, **40**A**2** are preferably driven in a particular sequence to simulate a flame while the LEDs **40**B may be driven to provide constant light. In embodiments, the control circuitry **44***a* may be used to control lighting of the UV light source **30** and the light sources **40**A**1**, **40**A**2** and **40**B even if they are not LEDs.

In embodiments, a power button or other input element **22** may be operable to activate the flickering effect of the LEDs **40**A**1**, **40**A**2** and/or the LEDs **40**B, or other light sources. In embodiments, the power button **22** may activate the fan **60** and/or the UV light source **30**, however, typically, the fan **60** and UV light source **30** will be activated together. In embodiments, the fan **60**, UV light source **30** and the flickering LEDs **40**A**1**, **40**A**2** will all be activated together by the power button **22**. In embodiments, the fan **60**, UV light source **30** and the LEDs **40**B may be activated together by the power button **22**. In embodiments, as noted above, the additional LEDs **40**B or other light sources may be used to provide for a constant light rather than flickering and may also be controlled by input from the power button **22**. In embodiments, the power button **22** may be positioned elsewhere. In embodiments, these constant light sources **40**B may be activated with or without activation of the fan **60** and/or the UV light source **30**. In embodiments, other input elements, other than the power button **22**, may be used to provide input to control the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and/or the constant light sources **40**B. In embodiments, a light sensor **31** may provide information regarding ambient light levels around the device **100**. In embodiments, the lighting element **40** may be activated when the ambient light level drops below a threshold level and may turn off when the light level rises above the threshold. In embodiments, the light sensor **31** may be a photocell, however, any suitable light sensor device may be used. In embodiments, the light sensor **31** may be provided on the top of the top portion **10***a*. In embodiments, the light sensor may **31** be integrated into a solar panel **18**. In embodiments, a separate light sensor **31** may be provided elsewhere on the lighting element **100**. In embodiments, the light sensor **31** may not be included.

FIG. **5** illustrates an exemplary block diagram of device **100**. In embodiments, the power source **80** provides power to the fan **60**, the UV light source **30** and the flickering LEDs **40**A**1**, **40**A**2** and LEDs **40**B. The control circuitry **44***a* on PCB **44** which may be or include the boost circuit, and/or other circuitry, may drive the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and LEDs **40**B. The control circuitry **44***a* may include other control circuitry to control activation of the fan **60**. As noted above, the flickering LEDs **40**A**1**, **40**A**2** may be driven in respective patterns to simulate the appearance of a flickering flame and the LEDs **40**B may be activated to provide constant light when desired. In embodiments, where a solar panel **18** is provided, the solar panel **18** may provide power to recharge the power source **80**. In embodiments, the solar panel **18** may not be included. In embodiments, the light portion **40** may include other charging circuitry or inputs to allow for USB or wireless charging, if desired. In embodiments, the control circuitry **44***a* may be connected to the power button **22** (or other input element) and may drive the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and constant light LEDs **40**B based on input provided. In embodiments, separate control circuitry may be provided and connected to the power button **22** and the control circuitry **44***a* to control the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and LEDs **40**B. In embodiments, the light level information provided by the light sensor **31**, when present, may be provided to the control circuitry **44***a*. In embodiments, the control circuitry **44***a* may include a processor, microprocessor or other control element or component to provide for control of the fan **60**, the UV light source **30** and the flickering LEDs **40**A**1**, **40**A**2**. In embodiments, control of the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and LEDs **40**B may be based on both input from the power button **22** and the light sensor. In embodiments, the power button **22** may be pressed once, or placed in a first position, to enter a light monitoring mode in which power is provided to one or more of the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** or constant light LEDs **40**B when the light information indicates a light level below a threshold. In embodiments, one or more of the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and the constant light LEDs **40**B may be deactivated when the light level rises above the threshold. In embodiments, pushing the button **22** again, or putting it in a second position, may directly activate one or more of the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and the constant light LEDs **40**B without consideration of the light level information. In embodiments, as noted above, each of the fan **60**, the UV light source **30**, the flickering LEDs **40**A**1**, **40**A**2** and constant light LEDs **40**B may be activated independently, if desired, based on operation of, or the position of, the power button and/or light level information provided by the light sensor. In embodiments, the fan **60** and UV light source **30** may be activated independent of the light level information. In embodiments, as noted above, other input elements may provide information to control the fan **60**, the UV light source **30**, the flickering LEDs **40A1**, **40A2** and the constant light LEDs **40B**. In embodiments, the device **100** may be placed in an off mode in which all of the fan **60**, the UV light source **30**, the flickering LEDs **40A1**, **40A2** and/or LEDs **40B** are deactivated and stay that way regardless of light sensor information until activation of the power button **22** or another input element.

Although the present invention has been described in relation to particular embodiments thereof, many other variations and modifications and other uses will become apparent to those skilled in the art. It is preferred, therefore, that the present invention be limited not by the specific disclosure herein.

What is claimed is:

1. An insect eliminating device comprising:

a body including a top portion and a bottom portion provided below the top portion;

a light portion mounted in the body and including:

a UV light element mounted in an open space provided in the top portion;

a flickering light portion mounted in the bottom portion;

a central conduit providing fluid communication between an open space in the top portion and the bottom portion of the body;

a fan, mounted in the bottom portion to direct air from the open space in the top portion, through the central conduit and into the lower portion; and

a grid mounted at a bottom end of the central conduit including a plurality of openings sized to allow air and insects to pass downward into the bottom portion of the base and prevent insects from passing upwards into the central conduit.

2. The insect eliminating device of claim **1**, wherein the UV light element is a UV light emitting diode.

3. The insect eliminating device of claim **1**, wherein the flickering light portion includes a first group of light emitting elements configured to emit light in a first pattern and a second group of light emitting elements configured to emit light in a second pattern.

4. The insect eliminating device of claim **3**, further comprising a support provided in the bottom portion of the body, wherein the first group of light emitting elements and the second group of light emitting elements are mounted on the support.

5. The insect eliminating device of claim **1**, further comprising a shade mounted in the bottom portion of the body, wherein the shade surrounds the flickering light portion.

6. The insect eliminating device of claim **1**, wherein the light portion includes at least one constant light element.

7. The insect eliminating device of claim **1**, further comprising a cage element provided around the bottom portion of the body configured to prevent insects from exiting the bottom portion of the body.

8. The insect eliminating device of claim **1**, further comprising a control circuit operably connected to the light portion and the fan to selectively activate the UV light portion, the flickering light portion and the fan to draw insects from the open space in the top portion into the central conduit and to the bottom portion of the body.

9. The insect eliminating device of claim **8**, further comprising an input device operably connected to the control circuit, wherein the control circuit selectively operates the light portion and the fan based on input provided by the input device.

10. The insect eliminating device of claim **1**, further comprising a light sensor operably connected to the control circuit, wherein the control circuit selectively activates the light portion and the fan based on an ambient light level detected by the light sensor.

11. The insect eliminating device of claim **10**, further comprising an input device operably connected to the control circuit, wherein the control circuit selectively activates the light portion and the fan based on input provided by the input device.

12. The insect eliminating device of claim **10**, further comprising a power source mounted in the body and configured to provide power to the control circuit, light portion and the fan.

13. The insect eliminating device of claim **12**, wherein the power source is a battery.

14. The insect eliminating device of claim **12**, wherein the power source is a rechargeable battery.

15. The insect eliminating device of claim **12**, further comprising a solar panel mounted on a top of the body and operably connected to the power source to recharge the power source.

16. The insect eliminating device of claim **12**, wherein the control circuit includes a charging circuit configured to control charging of the power source.

17. The insect eliminating device of claim **12**, further comprising a wire connectable to an exterior power source, wherein the wire is electrically connected to the power source and operable to recharge the power source.

18. The insect eliminating device of claim **1**, furth comprising a stand configured to receive the body.

19. The insect eliminating device of claim **1**, wherein the light portion further comprises a constant light source.

20. The insect eliminating device of claim **19**, wherein the constant light source includes a plurality of constant light elements.

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